

Marketing Campaign Analytics

Business Findings and Recommendations

Tools Used:

- PostgreSQL
- SQL
- Power BI

Dataset: Customer_360_Marketing_Campaign_Analytics

Table of Contents

1. Project Overview
2. Data Cleaning Summary
3. SQL Analysis Summary
4. Dashboard Overview
5. Key Business Findings
6. Business Recommendations
7. Conclusion

1. Project Overview

Objective

The objective of this project was to analyze customer demographics, purchasing behavior, product spending, and marketing campaign performance using PostgreSQL and Power BI. The project involved importing data into PostgreSQL, performing data cleaning and SQL analysis, and creating interactive dashboards in Power BI to generate meaningful business insights and support data-driven decision-making.

2. Data Cleaning Summary

The dataset was reviewed and cleaned before analysis to ensure data quality and accuracy.

Data Cleaning Activities Performed

- Imported customer dataset into PostgreSQL.
- Verified total records and columns.
- Checked for duplicate customer records.
- Checked missing values in Income.
- Replaced missing Income values with the average income.
- Identified and corrected negative spending values.
- Checked invalid birth years.
- Verified date consistency.
- Created calculated columns for business analysis:
 - Age
 - Age Group
 - Income Group
 - Total Children
 - Customer Value
 - Total Campaigns Accepted

3. SQL Analysis Summary

The following SQL analyses were performed on the dataset:

Customer KPIs

- Total Customers
- Total Spending
- Average Spending
- Average Income
- Total Purchases
- Campaign Response Rate
- Complaint Rate

Customer Analysis

- Spending by Age Group
- Spending by Income Group
- Spending by Education
- Spending by Marital Status
- Customer Value Analysis

Sales Analysis

- Product Spending

- Purchase Channel Analysis
- Country Analysis
- City Analysis
- Top Customers

4. Dashboard Overview

Two interactive dashboards were developed in Power BI.

Dashboard 1 – Executive Overview

This dashboard provides a high-level summary of customer performance using KPIs and visualizations.

Included Visuals

- Total Customers
- Total Spending
- Average Spending
- Average Income
- Total Purchases
- Campaign Response Rate
- Spending by Education
- Purchase Channel Distribution
- Income Group Analysis
- Customer Value Analysis
- Gender Distribution

Dashboard 2 – Customer & Campaign Analysis

This dashboard focuses on customer segmentation and campaign performance.

Included Visuals

- High Value Customers
- Complaint Rate
- Average Age
- Total Campaigns Accepted
- Spending by Age Group
- Spending by Education
- Spending by Marital Status

- Customer Value Distribution
- Customers by Country
- Top Cities by Spending
- Preferred Purchase Channel
- Customer Detail Table

5. Key Business Findings

Finding 1

High-income customers generated the highest overall annual spending and contributed the largest share of revenue.

Finding 2

Customers aged between 46 and 60 represented one of the strongest spending groups.

Finding 3

Web and Catalog channels were the most preferred purchasing channels.

Finding 4

Platinum and Gold customers generated significantly higher annual spending compared to Bronze customers.

Finding 5

Pakistan and USA contributed the highest overall customer spending within the dataset.

6. Business Recommendations

Recommendation 1

Develop exclusive marketing campaigns for high-income and high-value customers to maximize revenue.

Recommendation 2

Increase investment in Web and Catalog marketing channels since these channels generate a large proportion of customer purchases.

Recommendation 3

Strengthen loyalty programs for Bronze customers by offering personalized promotions and rewards to improve customer retention and encourage upgrades to higher customer segments.

7. Conclusion

This project successfully demonstrated a complete data analytics workflow using PostgreSQL, SQL, and Power BI. Data was imported, cleaned, transformed, analyzed, and visualized through interactive dashboards. The insights generated from this analysis can help businesses improve customer segmentation, optimize marketing strategies, and make informed business decisions.

Appendix

Project Tools

- PostgreSQL
- SQL
- Power BI Desktop

Dataset Information

- Dataset Name: Customer 360 Marketing Campaign Analytics
- Total Records: 1001
- Total Columns: 37